The turbulent boundary layer as an unconfined intermittent canopy: A structural synthesis

Guillem Borrell

Abstract

We develop a structural interpretation of the outer region of a zero-pressure-gradient turbulent boundary layer by combining three results that are usually discussed separately: the hierarchy of wall-attached momentum-transfer and vortex structures, the temporal similarity of those structures, and the geometry and dynamics of the turbulent/non-turbulent interface. The evidence supports an *intermittent-canopy* picture in which self-similar motions of many sizes overlap below the boundary-layer edge, while only the largest motions reach the intermittent region. Intense attached Reynolds-stress objects in channels have dimensions approximately proportional to their height and, at the identification threshold used in the source study, carry about 60% of the Reynolds stress while occupying less than 8% of a wall-parallel plane. Related vortex clusters have comparable aspect ratios and a scale-dependent population. Independent boundary-layer experiments and DNS identify analogous wall-attached packet and velocity structures through the logarithmic layer, although not at the T/NT interface itself. Time-resolved tracking shows that tall attached structures preserve this similarity in time: their lifetimes scale with their height and u_τ. They can therefore contribute to a canopy without introducing a new outer time scale that would break wall similarity. At the outer boundary, we adopt the interface definition and conditional coordinate established by Borrell and Jiménez [2016]. Their data show that the same wall-derived star scaling for vorticity and strain remains applicable through and well past the mean δ_{99}, including the intermittent excursions. This scaling continuity does not imply amplitude continuity: strain drops by more than one order of magnitude before the much weaker potential-flow branch decays exponentially. On the turbulent side, strain-normalised stretching and strain-eigenvalue statistics remain similar between the interface buffer and the core. We hypothesise that the interface is the convoluted outer envelope at which the wall-attached cluster hierarchy terminates: the intermittent region is where that hierarchy becomes sparse. The resulting cluster-to-interface connection is deliberately weak: attached structures are established in the cluster-bearing core, and interface-conditioned data show similar normalised physics up to the local turbulent edge, including its intermittent excursions, but the two have not been labelled jointly. This weak connection yields direct predictions for joint cluster/interface measurements.

Keywords: wall-bounded turbulence; turbulent boundary layer; coherent structures; attached eddies; turbulent/non-turbulent interface; intermittency

1 Introduction

Descriptions of zero-pressure-gradient turbulent boundary layers (ZPG TBLs) commonly use several complementary languages. Mean-flow descriptions distinguish viscous, logarithmic, wake and intermittent regions. Structural descriptions invoke streaks, sweeps, ejections, vortex clusters and attached eddies. Interface descriptions focus on the boundary separating rotational turbulence from the irrotational free stream [Corrsin and Kistler, 1955, Borrell and Jiménez, 2016]. These descriptions refer to different observables and cannot be identified with one another without an explicit argument.

Planar particle-image velocimetry established that the outer TBL contains repeated combinations of spanwise vorticity and strong Q2 motion [Adrian et al., 2000]. Spanwise scales grow linearly with wall distance through the logarithmic layer [Tomkins and Adrian, 2003], three-dimensional experiments recover vortex-packet signatures in an actual ZPG boundary layer [Dennis and Nickels, 2011], and boundary-layer DNS identifies self-similar wall-attached velocity structures up to approximately 0.18δ [Hwang and Sung, 2018]. Channel DNS provides the complementary gradient-based result that intense Reynolds-stress regions and vortex clusters include wall-attached, approximately self-similar families [del Álamo et al., 2006, Lozano-Durán et al., 2012]. Time-resolved DNS showed that the geometrical similarity of tall attached structures extends to their lifetimes [Lozano-Durán and Jiménez, 2014]. In parallel, conditional analysis of the turbulent/non-turbulent (T/NT) interface showed a sharp separation between turbulent-level gradients and a much weaker potential-flow strain branch [Borrell and Jiménez, 2016].

We ask whether these observations can be organised into one economical picture. The proposed *intermittent-canopy hypothesis* is:

> The logarithmic and outer regions contain a hierarchy of overlapping, approximately self-similar turbulent structures. Their population and spatial coverage decrease with height. Near the boundary-layer edge, the union of the surviving structures becomes sparse, and its convoluted vortical boundary is observed as the T/NT interface.

The term *canopy* conveys overlap, a distribution of heights and an irregular upper envelope. It does not imply a forest of isolated, permanent vortex tubes rooted at the wall. The structures extracted from DNS are multiscale regions rather than idealised individual vortices. In addition, most three-dimensional object evidence discussed below was obtained in channels, whereas the interface evidence was obtained in a spatially developing boundary layer. The synthesis is plausible only to the extent that the relevant logarithmic-layer organisation is shared by the two flows.

This article is a synthesis rather than a report of a new simulation. We consequently distinguish (i) observations displayed directly by published statistics, (ii) inferences supported by those observations, and (iii) the untested connection between the attached hierarchy and the interface. That distinction prevents a wall-attached Reynolds-stress object, a connected vortex cluster and a region enclosed by a T/NT interface from being treated as interchangeable.

2 Definitions and scales

Let x, y and z denote the streamwise, wall-normal and spanwise directions. The velocity fluctuations are (u, v, w), the friction velocity is u_τ, the kinematic viscosity is ν, and δ_{99} is the height at which the mean velocity reaches 99% of its free-stream value. Wall units are denoted by a superscript $+$, so that

$$\delta_{99}^+ = \frac{u_\tau \delta_{99}}{\nu}. \tag{1}$$

A quadrant object is a connected region in which the instantaneous tangential Reynolds stress exceeds a prescribed threshold. Ejections are Q2 events ($u < 0$, $v > 0$), and sweeps are Q4 events ($u > 0$, $v < 0$). In the object analysis of Lozano-Durán et al. [2012], an object is termed attached when its lowest point lies within approximately 20 wall units of the wall. This is an operational definition of geometric connectivity, not a statement about the birthplace of the full object.

Let $\omega = |\boldsymbol{\omega}|$ be the vorticity magnitude. We adopt without modification the vorticity-interface convention and interface-conditioned coordinate established by Borrell and Jiménez [2016]. Their dimensionless interface scales are

$$\omega^* = \omega^+ \sqrt{\delta_{99}^+}, \qquad S^* = S^+ \sqrt{\delta_{99}^+}, \tag{2}$$

where $\omega^+ = \omega\nu/u_\tau^2$ and $S^+ = S\nu/u_\tau^2$. The corresponding dimensional edge-vorticity scale is

$$\omega_e \sim \frac{u_\tau^2}{\nu\sqrt{\delta_{99}^+}} = \sqrt{\frac{u_\tau^3}{\nu\delta_{99}}}, \tag{3}$$

up to constants. It follows from the local-equilibrium estimate $\nu\omega_e^2 \sim u_\tau^3/\delta_{99}$. Note that $\sqrt{\delta_{99}^+}$, rather than its inverse, appears in the definition of ω^*.

With the interface convention fixed as in Borrell and Jiménez [2016], let

$$\gamma(y) = \Pr\{\text{point at height } y \text{ lies on the turbulent side}\}. \tag{4}$$

This is not generally equal to $\Pr\{h_k > y\}$, where h_k is a cluster height. Pointwise coverage also depends on each object's footprint, overlap and connectivity. A cluster-height cumulative distribution is therefore not, by itself, an intermittency profile.

3 A self-similar structural hierarchy

3.1 Three-dimensional attached objects

The vortex-cluster analysis of del Álamo et al. [2006] separates compact detached objects from tall attached objects. Rather than using one arbitrarily selected cluster as evidence for the hierarchy, figure 1 shows the population statistics directly. The Cartesian bounding dimensions scale approximately as

$$\Delta_x \simeq 3\Delta_y, \qquad \Delta_z \simeq 1.5\Delta_y. \tag{5}$$

The data in figure 1 are from the S1900 channel at $Re_\tau \simeq 1900$, comparable to the upper friction-Reynolds-number range of the boundary-layer interface data discussed in section 4. Reynolds number is therefore not the main mismatch between the two datasets; flow geometry and object definition are. Figure 1 supplies the wall-similar cluster hierarchy required by the hypothesis; the interface side of the proposed identification is developed in section 4.

The number density per unit wall area and height has a power-law tail. As the identification threshold approaches the percolation transition, its exponent approaches -3, as expected for a Townsend-like geometrically self-similar population [Townsend, 1976, del Álamo et al., 2006].

Lozano-Durán et al. [2012] performed a corresponding three-dimensional analysis of intense momentum-transfer events. Their objects separate into attached and detached populations (figure 2). At their selected threshold, attached Q2 and Q4 objects occupy less than 8% of a wall-parallel plane but account for approximately 60% of the mean Reynolds shear stress at all heights. This threshold-specific statement is more defensible than claiming that such objects fill the flow or carry nearly all of the stress.

The attached Q family has dimensions

$$\Delta_x \simeq 3\Delta_y, \qquad \Delta_z \simeq \Delta_y, \tag{6}$$

except for the largest objects crossing the channel centreline. Figure 3 displays the distributions underlying equation (6). The similarity between equations (5) and (6) is significant, but it does not imply that the vortex and momentum-transfer objects are identical. More than 80% of attached vortex clusters intersect a Q object, whereas the reverse probability decreases with height because tall Q objects outnumber tall clusters [Lozano-Durán et al., 2012].

The preferred arrangement is a side-by-side Q2–Q4 pair with a vortex cluster associated with the Q2 and with the shear layer below the Q4. The conditional and instantaneous fields in figure 12 of Lozano-Durán et al. [2012] show that the instantaneous object is less regular than its average. For the canopy hypothesis, the important point is not its exact shape but its function:

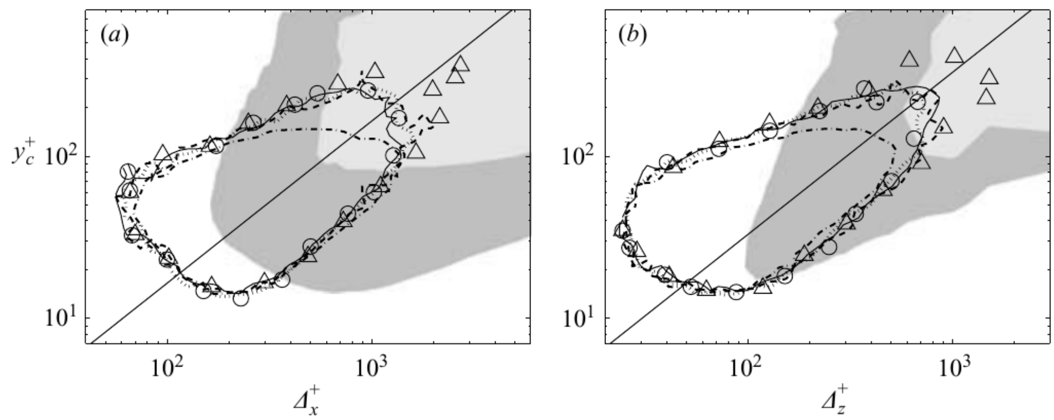

Figure 1: Joint distributions of attached-cluster streamwise (a) and spanwise (b) dimensions versus the wall distance of their centres. The diagonal trends are the direct evidence for geometrical self-similarity. Data are from the S1900 channel at $Re_\tau \simeq 1900$. Reproduced from figure 7 of del Álamo et al. [2006].

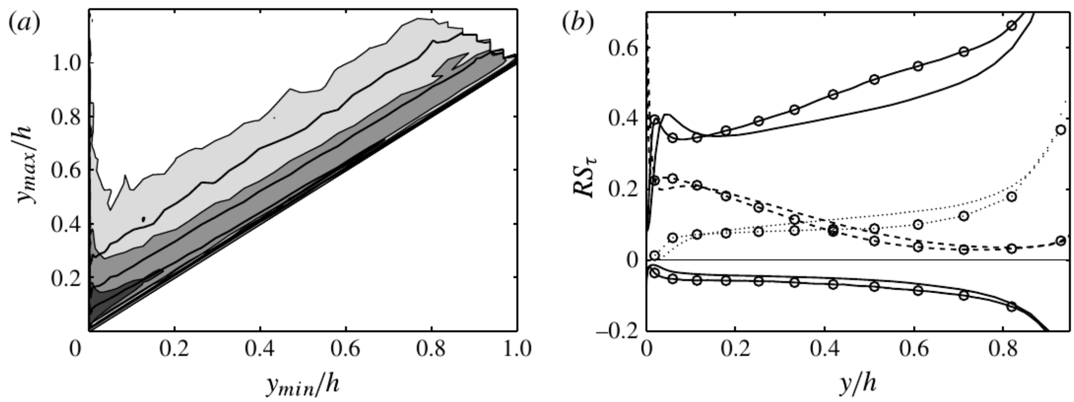

Figure 2: Left: joint distribution of the lowest and highest wall distances of identified Q objects. Right: fractional contributions to Reynolds stress from attached Q2 and Q4 objects, detached objects and counter-gradient events. Reproduced from figure 2 of Lozano-Durán et al. [2012].

the Q2 ejection supplies the outward-transport component linked to the vortical object. The cluster–ejection pair is therefore a candidate dynamical element of the height-scaled hierarchy.

The channel result is not the only evidence for an attached hierarchy. In ZPG boundary layers, Tomkins and Adrian [2003] found spanwise length scales growing linearly with wall distance, Dennis and Nickels [2011] measured three-dimensional vortex-packet signatures, and Hwang and Sung [2018] identified self-similar attached structures in all three velocity components. The last study reconstructed logarithmic intensity from the attached population over $100 < y^+ < 0.18\delta^+$. These studies concern velocity or planar vortex signatures rather than the connected gradient clusters used here, and they stop below the intermittent edge. Nevertheless, they anchor the wall-attached hierarchy in a boundary layer at approximately 0.2δ, rather than only in channels.

Together, these observations provide the proposed canopy with candidate building blocks and a transport mechanism. The height-scaled attached objects supply the hierarchy; their association with Q2 ejections supplies outward transport; and their decreasing population with height supplies the sparsity required near the edge. They do not make velocity, stress and vorticity objects interchangeable.

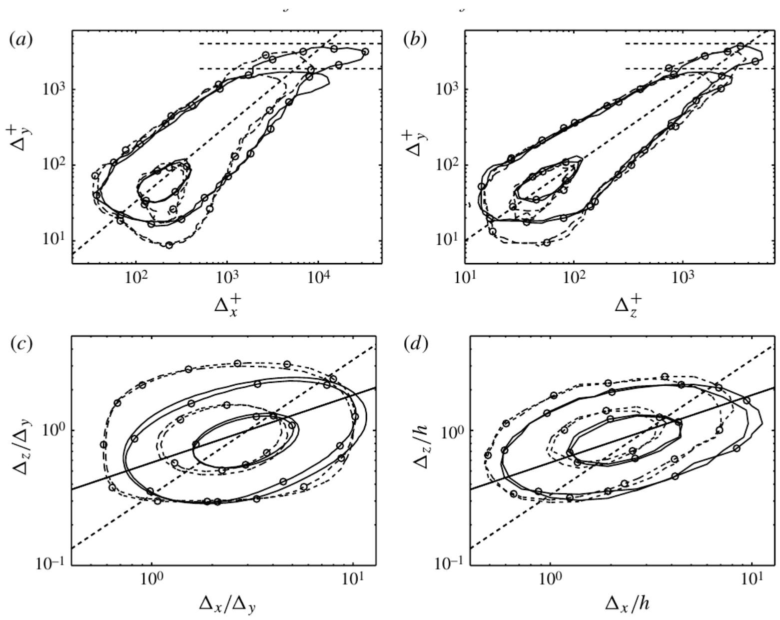

Figure 3: Candidate building blocks of the canopy. The diagonal trends in panels (a,b) show that attached Q2 and Q4 dimensions scale with object height. Panel (d) shows that the largest thresholded objects depart from this simple family and should be interpreted as composites rather than as single self-similar eddies. Reproduced from figure 5 of Lozano-Durán et al. [2012].

3.2 From self-similar elements to a canopy

The canopy hypothesis does not require the outer layer to be one giant coherent object. In fact, simple self-similarity breaks down for the largest thresholded channel structures. Objects crossing the centreline account for roughly one third of the stress carried by attached Q objects; fewer than 1% of all identified objects account for about 60% of the attached volume; and streamwise lengths can approach 20δ [Lozano-Durán et al., 2012]. Figure 4 shows that such a very-large Q2 is assembled from smaller pieces.

This composite character is part of the canopy argument rather than an exception to it. The interface would be the envelope of the union of many overlapping height-scaled elements; it need not coincide with the boundary of a single segmented cluster. Consequently, the height of a percolated object is not the size of one dynamical eddy. Long streamwise-velocity correlations and global spectral modes can also combine active and inactive motions and should not be identified with the rotational canopy itself [del Álamo et al., 2004].

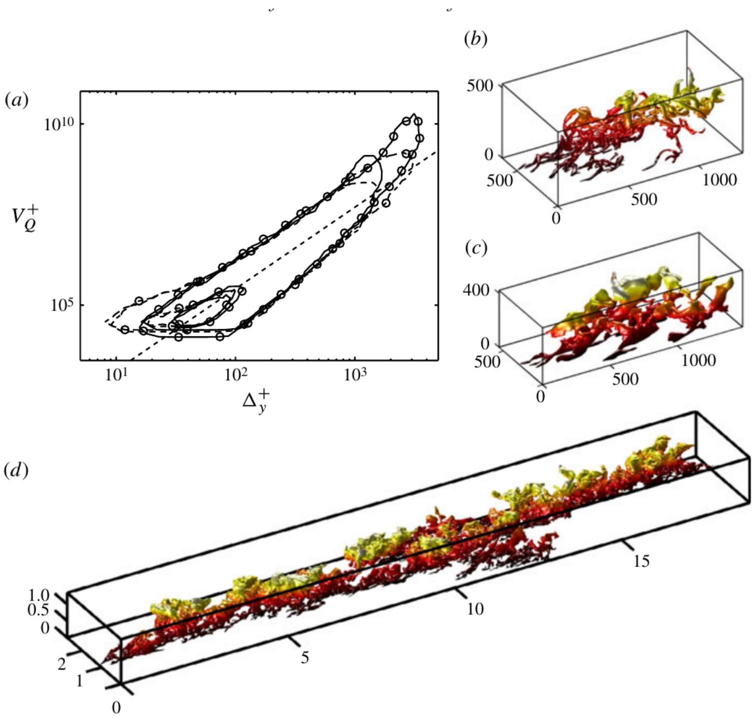

Figure 4: From elements to canopy. Panels (b,c) show an attached vortex cluster and an attached Q2; panel (d) shows a very-large centreline-crossing Q2 assembled from smaller pieces. The canopy is interpreted as the sparse union of such height-scaled elements, not as one monolithic eddy. Reproduced from figure 8 of Lozano-Durán et al. [2012].

3.3 Preservation of wall similarity

The time-resolved result needed by the cluster-to-interface hypothesis is whether the structures forming the canopy preserve wall similarity. Lozano-Durán and Jiménez [2014] found that tracked tall attached branches remain geometrically self-similar and that their lifetimes scale as

$$\frac{Tu_\tau}{\Delta_y} = O(1). \tag{7}$$

The tracked structures therefore introduce neither a separate outer length nor a separate outer time scale: their dimensions scale with Δ_y, and their duration with Δ_y/u_τ. This is the role of the time-resolved evidence in the hypothesis: the coherent structures can form a canopy while preserving wall similarity. The identification of that canopy with the boundary-layer interface is then supplied by the termination and inner-side evidence in section 4.

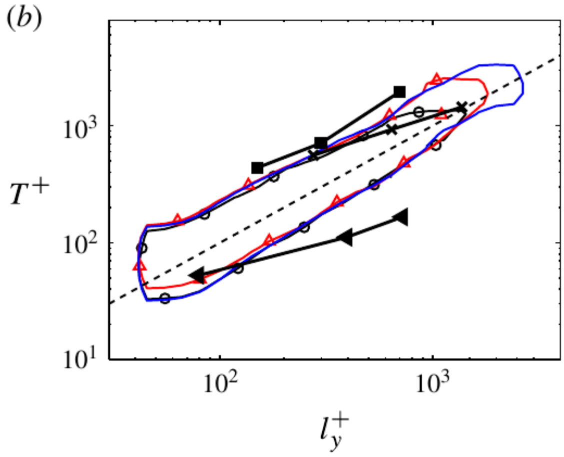

Figure 5: Lifetime of tall attached Q branches versus branch height. The dashed diagonal is $T^+ = l_y^+$. The temporal scale is set by the same height and friction velocity that describe the attached hierarchy. Cropped from figure 9(b) of Lozano-Durán and Jiménez [2014].

4 Wall-similar physics at the established T/NT interface

We use the interface-conditioned statistics reported by Borrell and Jiménez [2016] directly. Because their coordinate follows the local interface, the conditional ensemble includes both valleys below and bulges above the mean δ_{99}. Figure 6 records the relevant scaling observation; the amplitude distinction between turbulent and potential strain is noted in its caption.

4.1 Core-like organisation up to the turbulent side

Figure 7 concerns the turbulent side of the interface, not continuity into the potential flow. Its key conclusion is that vorticity in the interface buffer lives in essentially the same straining environment as the turbulent core. Panel (a) follows the joint vorticity and strain distributions towards the interface, and panel (b) follows vortex stretching and compression. In panels (c,d), the samples at 7η and 100η correspond respectively to the interface buffer and the turbulent core. Once normalised by the local strain magnitude, the stretching/compression and strain-eigenvalue distributions are similar. The internal organisation therefore remains core-like as the cluster hierarchy approaches its outer termination, even though the amplitudes subsequently collapse across the interface.

This is the weak bridge used in the present hypothesis. Independent boundary-layer evidence places a wall-attached hierarchy in the cluster-bearing flow up to approximately $0.2\delta_{99}$, while the interface-conditioned statistics show essentially the same normalised straining physics from the core to the local turbulent edge near δ_{99}, including the ensemble of peaks above it. We infer—but do not directly observe—that the attached cluster hierarchy carries that physics to the edge and terminates there. The T/NT interface is proposed to be the envelope of those terminations.

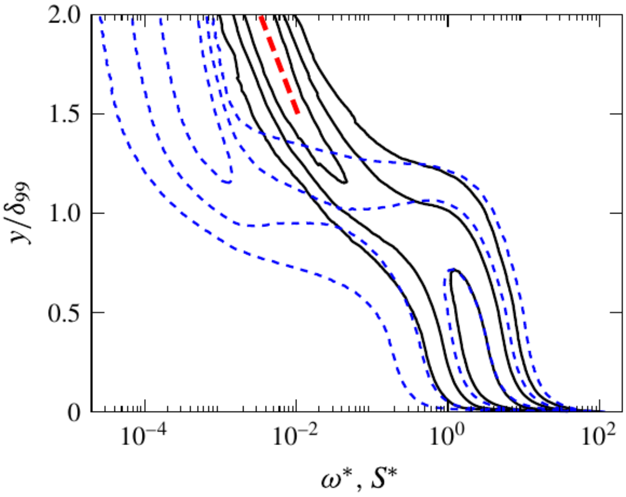

Figure 6: Joint wall-normal distributions in the same wall-derived star units: vorticity ω^* (blue dashed contours) and strain S^* (black solid contours). The relevance to the hypothesis is that this scaling remains applicable through and well past the mean δ_{99}, including the intermittent excursions sampled by the local interface. This is scaling continuity, not amplitude continuity: turbulent-level strain drops by more than one order of magnitude near the edge, after which the much weaker potential branch follows the exponential decay marked in red. Reproduced from figure 20 of Borrell and Jiménez [2016].

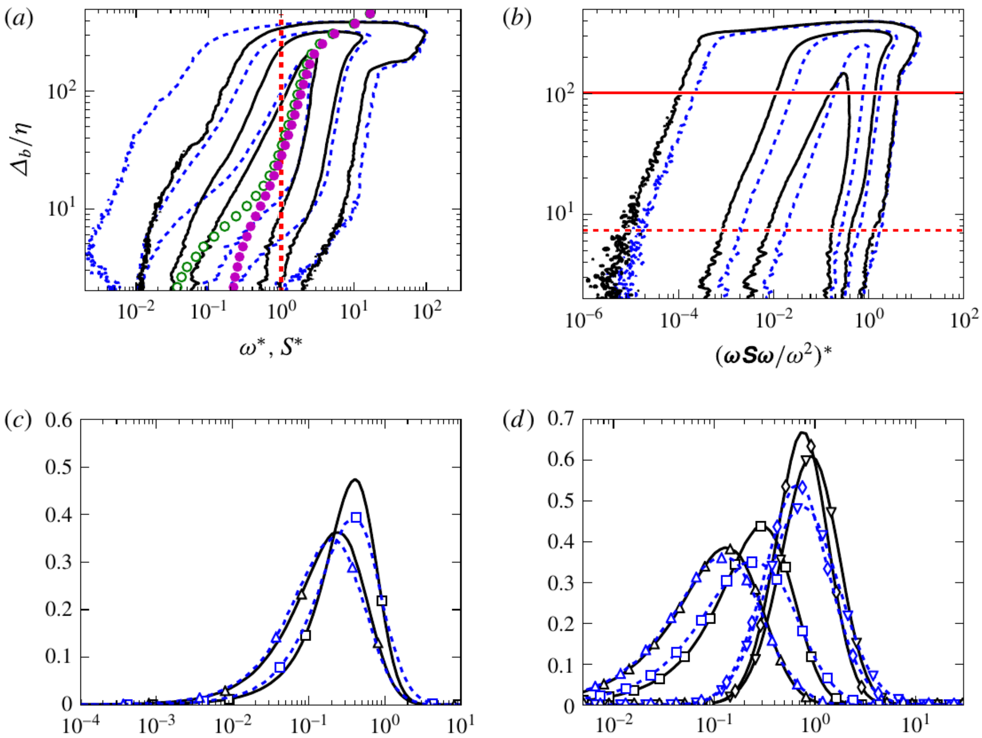

Figure 7: Conditional gradient statistics on the turbulent side of the vorticity interface. Blue dashed curves sample the interface buffer at 7η; black solid curves sample the core at 100η. Their near collapse after local-strain normalisation shows persistence up to the edge, not continuity across it. Reproduced from figure 23 of Borrell and Jiménez [2016].

5 Comparative evidence: channel global mode versus open canopy

A channel supplies a quantitative counterfactual for the canopy hypothesis. Jiménez et al. [2010] show that large-scale velocity correlations cross the channel centreline and that ejections from one half penetrate the other. The largest attached motions can therefore be incorporated into a global centre-channel organisation. A TBL has no opposing turbulent half-channel: its attached population instead approaches an open intermittent boundary. Figure 8 documents the centreline coupling that distinguishes the internal flow.

Higher-Reynolds-number one-point statistics show that all three velocity-component intensities exceed their channel values in the TBL, with the largest difference near $y/\delta \simeq 0.35$–0.5 and little movement of that location over the available Reynolds-number range [Sillero et al., 2013]. This is consistent with an edge-mediated pressure and entrainment effect, not with a dynamically empty outer TBL.

Three-dimensional correlations quantify the difference [Sillero et al., 2014]. Weak streamwise-velocity correlations reach approximately 18δ in channels but only approximately 7δ in boundary layers, even though their cross-flow organisation is qualitatively similar. Figure 9 shows that the channel streamwise correlation length continues to grow to $y/\delta \simeq 0.5$, while the boundary-layer value levels off much earlier. The source study argues that intermittency prevents the corresponding global channel structure from forming in the TBL.

The defensible comparison is therefore:

- a channel has no T/NT interface, communicates across its centreline and supports very long global streamwise modes;
- a boundary layer has an open intermittent edge, stronger pressure and transverse-velocity fluctuations, shorter streamwise correlations and a weak potential-flow response beyond the vorticity boundary.

This difference supports the canopy hypothesis in a limited but relevant way. In a channel, the largest attached motions can be subsumed into a global centreline mode. In a boundary layer, that competing organisation is absent: the attached population can instead become sparse and terminate at the open T/NT interface. Correlations do not identify vortex clusters, so this is comparative support rather than a direct cluster measurement.

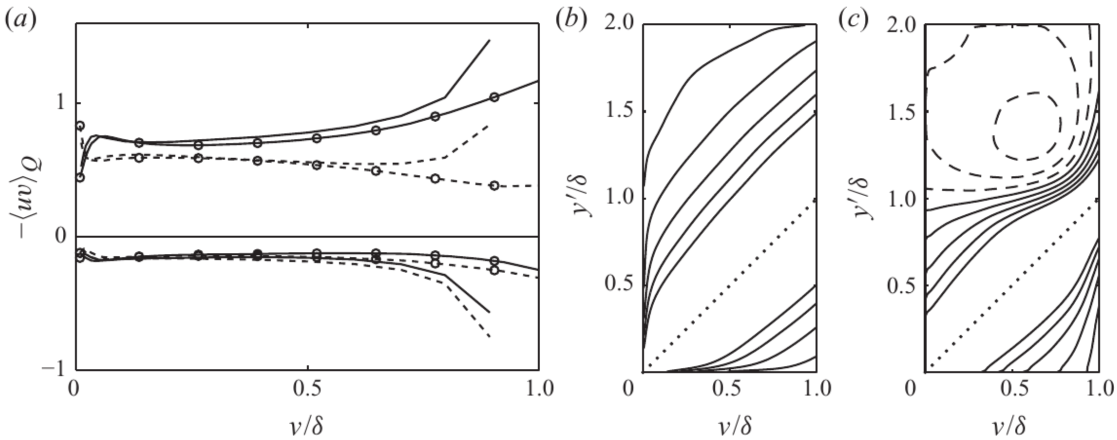

Figure 8: Why the channel is a useful counterfactual. Panel (a) compares quadrant contributions; panels (b,c) show large-scale channel correlations crossing $y/\delta = 1$. In a channel, outer attached motions enter a global centreline organisation rather than terminating at an open intermittent canopy. Reproduced from figure 5 of Jiménez et al. [2010].

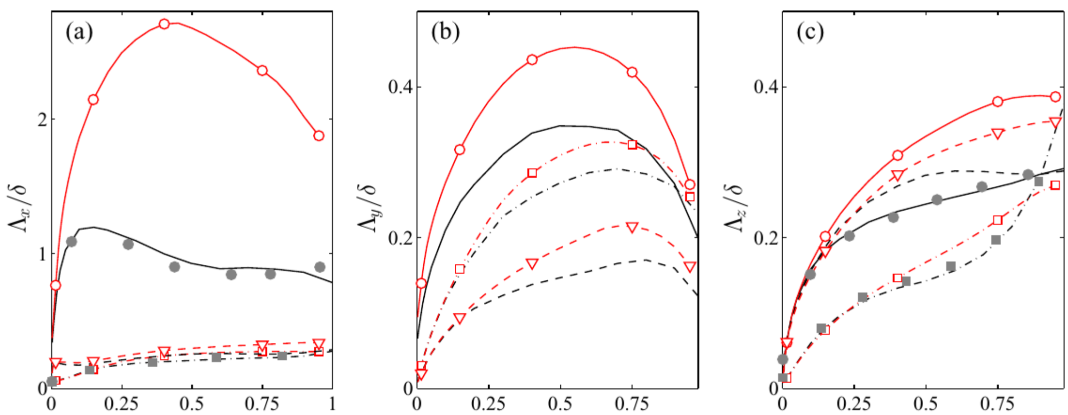

Figure 9: Quantitative evidence that the centre-channel mode is not reproduced in the boundary layer. In panel (a), the red channel streamwise correlation continues to grow to $y/\delta \simeq 0.5$, whereas the black boundary-layer curve levels off earlier; panels (b,c) show the shorter transverse scales. Reproduced from figure 10 of Sillero et al. [2014].

6 A weak cluster-to-interface connection

The argument deliberately combines strong observations through one weak inference.

1. In channels, attached vortex and momentum-transfer clusters form a wall-similar hierarchy, are associated with ejections, and preserve height-based length, velocity and time scaling to the outermost wall distances.
2. The channel/boundary-layer comparison shows that the global centre-channel mode is an internal-flow organisation that is absent from the open intermittent boundary layer, allowing the attached population to thin into a canopy rather than being subsumed at a centreline.
3. Independent ZPG-boundary-layer measurements and DNS recover wall-attached packet and velocity structures through the logarithmic layer, reaching approximately $0.2\delta_{99}$.
4. Interface-conditioned statistics include the local intermittent valleys and peaks and show that the normalised straining environment in the interface buffer is essentially the same as in the turbulent core.
5. The same wall-derived scaling for ω and S remains applicable through the local edge and its excursions beyond δ_{99}, even though turbulent-level amplitudes terminate sharply and the exterior branch is much weaker and potential.

The weak inference is that the attached gradient clusters known in channels are the carriers of the wall-similar physics observed in the boundary-layer core and remain its carriers up to the local interface, including canopy peaks above δ_{99}. On that basis, we hypothesise that the intermittent region is the sparse upper support of the wall-attached hierarchy and that the T/NT interface is its convoluted outer envelope. The inference is weak because the channel clusters, boundary-layer attached velocity structures and interface-conditioned gradients are different observables from different datasets; it is nevertheless constrained by their common scaling and dynamics.

A joint object/interface analysis provides a direct test: vorticity clusters, Q structures and the interface should be extracted simultaneously from the same TBL snapshots and their conditional intersections measured. The observation that more than 80% of channel vortex clusters intersect a Q object supplies a concrete expectation for that test.

A similarly careful statement is required for vorticity. For incompressible, constant-density flow,

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u} + \nu\nabla^2\boldsymbol{\omega}. \tag{8}$$

There is no baroclinic source in equation (8). The wall supplies vorticity, stretching and tilting redistribute and amplify it, and viscosity diffuses it across the T/NT interface. This global statement does not imply a Lagrangian conveyor belt in which identifiable vortex objects travel intact from the wall to δ_{99}. The tracking evidence argues against that literal interpretation.

7 Direct tests of the hypothesis

The canopy interpretation can be tested quantitatively using existing DNS fields or modest extensions of them.

7.1 Joint object/interface labelling

For each snapshot, identify the T/NT interface with the established convention of Borrell and Jiménez [2016], together with attached vortex clusters and attached Q objects. Measure

$$P(d_{\mathrm{TNTI}} < a\lambda \mid \text{top of attached object}) \tag{9}$$

and compare it with randomised objects of the same size and height. A canopy interpretation predicts an excess near the tops of the largest objects.

7.2 Coverage rather than height alone

Construct the pointwise union of objects and compare its coverage

$$\gamma_Q(y; H) = \Pr\{(x, y, z) \in \cup_k Q_k(H)\} \tag{10}$$

with the vorticity intermittency in equation (4). Object footprints and overlaps must be retained; a cumulative height distribution is insufficient.

7.3 Conditional budgets at structural tips

Evaluate mean-shear production, nonlinear transport, vortex stretching and viscous diffusion conditioned jointly on distance from the interface and distance from an object top. If sparse object tips form the interface, these budgets should differ from those at generic interface points of the same height. The test would determine whether the core-like straining on the turbulent side in figure 7 is specifically associated with attached cluster tips.

7.4 Matched internal/external comparison

Use identical thresholds and object definitions at matched Re_τ. Compare object populations, height scaling, centreline or interface termination and canopy coverage. This would replace qualitative collision and freedom metaphors with measurable differences.

8 Conclusions

The published evidence supports a unified but qualified structural narrative.

1. **Hierarchy.** Three-dimensional vortex clusters and intense Reynolds-stress objects contain attached, approximately self-similar families whose wall-parallel dimensions scale with their height. At the threshold of Lozano-Durán et al. [2012], attached Q objects carry about 60% of the Reynolds stress while covering less than 8% of a plane.
2. **Wall similarity.** Tracked tall attached structures preserve geometrical and temporal similarity: their dimensions scale with height and their lifetimes with Δ_y/u_τ. Independent boundary-layer studies recover analogous attached organisation to approximately $0.2\delta_{99}$.
3. **Wall scaling at the edge.** The same wall-derived star scaling for vorticity and strain remains applicable through and beyond the mean δ_{99}. Core-like normalised straining persists up to the turbulent side, even though turbulent-level amplitudes terminate sharply and only a weak potential branch remains outside.
4. **Comparative support.** Channels develop a global centreline mode with streamwise correlations much longer than those in boundary layers. Its absence from the TBL removes a competing outer organisation and allows the attached hierarchy to thin towards an open canopy.
5. **Weak cluster-to-interface connection.** Because the interface buffer has essentially the same normalised straining environment as the core, we infer that the wall-attached hierarchy carries its wall-similar physics from approximately $0.2\delta_{99}$ to the local interface, including peaks above δ_{99}. The T/NT interface is proposed to be the envelope where that sparse hierarchy terminates.

The main value of the canopy framework is not that it abolishes the outer layer. It proposes a weak, testable identification between the wall-similar attached hierarchy and the structural support of the T/NT interface. Figures 6 and 7 constrain that identification by showing respectively that wall-derived scaling remains applicable across the intermittent excursions, despite the amplitude separation at the edge, and that the normalised organisation remains core-like up to the turbulent side.

Data availability

The DNS statistics and several datasets associated with the cited studies are available through the Fluid Dynamics Group at Universidad Politécnica de Madrid: https://torroja.dmt.upm.es/, https://torroja.dmt.upm.es/turbdata/ and https://torroja.dmt.upm.es/paperdat/. Availability differs by paper.

Figure provenance

References

Ronald J. Adrian, Carl D. Meinhart, and Christopher D. Tomkins. Vortex organization in the outer region of the turbulent boundary layer. *Journal of Fluid Mechanics*, 422:1–54, 2000. doi: 10.1017/S0022112000001580.

Guillem Borrell and Javier Jiménez. Properties of the turbulent/non-turbulent interface in boundary layers. *Journal of Fluid Mechanics*, 801:554–596, 2016. doi: 10.1017/jfm.2016.430.

Stanley Corrsin and Alan L. Kistler. Free-stream boundaries of turbulent flows. Technical Report NACA-TR-1244, National Advisory Committee for Aeronautics, 1955. URL https://ntrs.nasa.gov/citations/19930091020.

Juan C. del Álamo, Javier Jiménez, Paulo Zandonade, and Robert D. Moser. Scaling of the energy spectra of turbulent channels. *Journal of Fluid Mechanics*, 500:135–144, 2004. doi: 10.1017/S002211200300733X.

Juan C. del Álamo, Javier Jiménez, Paulo Zandonade, and Robert D. Moser. Self-similar vortex clusters in the turbulent logarithmic region. *Journal of Fluid Mechanics*, 561:329–358, 2006. doi: 10.1017/S0022112006000814.

David J. C. Dennis and Timothy B. Nickels. Experimental measurement of large-scale three-dimensional structures in a turbulent boundary layer. part 1. vortex packets. *Journal of Fluid Mechanics*, 673:180–217, 2011. doi: 10.1017/S0022112010006324.

Jinyul Hwang and Hyung Jin Sung. Wall-attached structures of velocity fluctuations in a turbulent boundary layer. *Journal of Fluid Mechanics*, 856:958–983, 2018. doi: 10.1017/jfm.2018.727.

Javier Jiménez, Sergio Hoyas, Mark P. Simens, and Yoshinori Mizuno. Turbulent boundary layers and channels at moderate Reynolds numbers. *Journal of Fluid Mechanics*, 657:335–360, 2010. doi: 10.1017/S0022112010001370.

Adrián Lozano-Durán and Javier Jiménez. Time-resolved evolution of coherent structures in turbulent channels: characterization of eddies and cascades. *Journal of Fluid Mechanics*, 759:432–471, 2014. doi: 10.1017/jfm.2014.575.

Adrián Lozano-Durán, Oscar Flores, and Javier Jiménez. The three-dimensional structure of momentum transfer in turbulent channels. *Journal of Fluid Mechanics*, 694:100–130, 2012. doi: 10.1017/jfm.2011.524.

Juan A. Sillero, Javier Jiménez, and Robert D. Moser. One-point statistics for turbulent wall-bounded flows at Reynolds numbers up to $\delta^+ \approx 2000$. *Physics of Fluids*, 25(10):105102, 2013. doi: 10.1063/1.4823831.

Juan A. Sillero, Javier Jiménez, and Robert D. Moser. Two-point statistics for turbulent boundary layers and channels at Reynolds numbers up to $\delta^+ \approx 2000$. *Physics of Fluids*, 26(10): 105109, 2014. doi: 10.1063/1.4899259.

Christopher D. Tomkins and Ronald J. Adrian. Spanwise structure and scale growth in turbulent boundary layers. *Journal of Fluid Mechanics*, 490:37–74, 2003. doi: 10.1017/S0022112003005251.

A. A. Townsend. *The Structure of Turbulent Shear Flow*. Cambridge University Press, Cambridge, 2 edition, 1976.